C S3081 Mini Project Report

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GitHub: https://github.com/aaryansinhaa/cs3081-command-parser

Task 1: k3d cc3081, SeaweedFS, 3000 teacher labels, 2810 kept, checksum stable.

Task 2: Pythia-70m student v2 99pct JSON validity, 46pct full_frame on validation.

Task 3: Image 1.66GB, FAULT_INJECT 500, 50 docker requests peak 403MiB no OOM.

Task 4: Ingress /parse, cold start median 12s, rollout v1 to v2 and undo.

Task 5: subnets.md and nacl.json, nacl_sim 6/6.

Task 6: pytest 11 passed, gitleaks clean, CI workflow in repo.

Task 7: loadtest p50 2800ms p95 7900ms rps 1.49, cost 1.47 INR per 1k requests.

Limitation: single k3d node; load errors during CPU-heavy rollouts documented.

See results/ and report/report.md for tables and CSV evidence.